MILITARY SPECIFICATION

GRAIN, IGNITER

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the igniter grain critical item, herein referred to as "grain."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1337

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MILITARY

MIL-M-382	Magnesium Powder (For Use in Ammunition).
MIL-I-85326	Igniter, Rocket Motor, MARK 297 MOD 0.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing.
MIL-STD-831	Test Reports, Preparation of.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

OTHER GOVERNMENT DOCUMENTS

Code of Federal Regulations (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C 20402.)

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1555641	Grain.
1568376	Igniter, Rocket Motor MK 264 MOD 1 Assembly.
1204AS201	Igniter, Rocket Motor, MARK 297 MOD 0.

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PUBLICATIONS

Naval Sea Systems Command
(Code Ident 53711)

WS 3853	Igniter, Rocket Motor, MARK 264 MOD 1.
WS 7682	Copolymer, Vinylidene Difluoride and Hexafluoropropylene.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM D 792	Specific Gravity and Density of Plastics by Displacement.
ASTM D 1457	PTFE Molding and Extrusion Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The grain is a pyrotechnic component composed mainly of magnesium and polytetrafluoroethylene (PTFE) extruded under pressure into a cruciform shape. The grain is used in rocket motor igniters.

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3.2 Characteristics.

3.2.1 Function.

3.2.1.1 MARK 264 igniter. The grains, when installed in the MARK 264 igniter, shall meet the performance requirements of WS-3853.

3.2.1.2 MARK 297 igniter. The grains, when installed in the MARK 297 igniter, shall meet the performance requirements of MIL-I-85326.

3.2.2 Density. The density of the grain shall be 1.86 ±0.02 grams per cubic centimeter.

3.2.3 Composition. The grain shall be composed of the ingredients in the amounts as specified in Table I.

TABLE I. Composition.

Ingredients	Applicable document	Parts by weight	
		Minimum	Maximum
Magnesium powder, Type III, granulation No. 16 Magnesium powder, Type III, granulation 325	MIL-M-382 1/	52	56
Plastic molding material PTFE	ASTM D 1457 Type IV	14	16
Plastic molding material PTFE	See Table II 2/	14	16
Copolymer, vinylidene fluoride and hexafluoropropylene	WS 7682	15	17

1/ The ratio of magnesium powder granulation No. 16 to granulation No. 325 mesh may be varied from a nominal ratio of 50:50 by weight in order to control the burning rate characteristics of the propellant.

2/ The plastic molding material PTFE shall conform to all the requirements of ASTM D 1457 except as shown in Table II.

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TABLE II. Plastic molding material composition.

Requirements	Minimum	Maximum
For resin:		
Bulk density, g/liter	575	675
Particle size, average diameter, micrometer	475	625
Water content, percent	---	0.03
Melting peak temperature, °C	317	337
Extrusion pressure	N/A	N/A
For molded specimens:		
Thermal instability index	---	50
Standard specific gravity	213	218
Tensile strength, MPa	13.8	---
Elongation, percent	140	---

3.2.4 Heat of explosion. The heat of explosion shall be 1,450 ±100 calories per gram.

3.2.5 Burning rate. The grain burning rate shall be as specified on Figure 1. The burning rate curve, as determined by the method of least squares, shall fall within the acceptable range as specified on Figure 1. All burning rate data shall be used in determining the burning rate curve.

3.3 Design and construction.

3.3.1 Production drawings. The grain shall be manufactured in accordance with Drawing 1555641 and as specified herein.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2.3).

3.5 Workmanship. The grain shall be manufactured in a manner to assure conformance with all the requirements of this specification and shall be uniform and free from all foreign materials or contaminants.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system of quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Quality conformance inspection (see 4.3).

4.2 First article inspection. When specified in the contract or purchase order (see 6.2.3), a first article sample of 70 grains shall be delivered to the testing activity designated in the contract or purchase order. The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample shall be at the risk of the contractor.

4.2.1 First article sample. The first article sample shall be subjected to the tests and inspections of Table III.

4.2.1.1 First article acceptance criteria. Failure of any grain sample to meet the requirements specified herein, when subjected to the tests and inspections of Table III, shall cause rejection of the first article sample.

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TABLE III. First article inspections.

Inspections	Requirement	Method
MARK 264 igniter function	3.2.1.1	4.4.1.1
or		
MARK 297 igniter function	3.2.1.2	4.4.1.2
Density	3.2.2	4.4.2
Composition	3.2.3	4.4.3
Heat of explosion	3.2.4	4.4.4
Burning rate	3.2.5	4.4.5
Workmanship	3.5	4.5

4.2.1.2 Test reports. When specified in the contract or purchase order (see 6.2.2), test reports, in accordance with MIL-STD-831, shall be prepared by the contractor for the contracting activity for all first article inspections.

4.3 Quality conformance inspection. Each production lot shall be subjected to the tests and inspections of Table IV.

TABLE IV. Quality conformance inspections.

Inspections	Requirement	Method
Density	3.2.2	4.4.2
Composition	3.2.3	4.4.3
Heat of explosion	3.2.4	4.4.4
Burning rate	3.2.5	4.4.5
Workmanship	3.5	4.5

4.3.1 Production lot. A production lot shall consist of all grains from one batch of materials extruded in a continuous process, using the same procedures, equipment, and conditions.

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4.3.2 Sampling. Unless otherwise specified herein, sampling shall be in accordance with MIL-STD-105.

4.3.3 Quality conformance acceptance criteria. Failure of any production sample to meet the requirements as specified herein, when subjected to the tests of Table IV, shall cause rejection of the production sample.

4.4 Test methods. Tests shall be performed using the test apparatus, ingredients, and procedures as specified herein.

4.4.1 Function tests.

4.4.1.1 MARK 264 igniter. Twenty-four grains shall be loaded into igniters in accordance with Drawing 1568376. The loaded igniters shall be temperature conditioned and fired in accordance with the function tests of WS-3853. The grains shall meet the requirements of 3.2.1.1.

4.4.1.2 MARK 297 igniter. Twenty-four grains shall be loaded into igniters in accordance with Drawing 1204AS201. The loaded igniters shall be temperature conditioned and fired in accordance with the arm function tests of MIL-I-85326. The grains shall meet the requirements of 3.2.1.2.

4.4.2 Density. Ten grain samples shall be subjected to the density test of ASTM D 792, except that the immersion agent shall be hexane. The density of each grain sample shall meet the requirements of 3.2.2.

4.4.3 Composition. Composition of the grain shall be determined as follows:

a. Weigh three 1-gram samples of the propellant to the nearest milligram. Digest each 1-gram sample in a 250-milliliter (ml) beaker with 75 ml reagent-grade acetone for not less than 16 hours.

b. Filter the sample through a weighed fine crucible. Acetone shall be used to transfer the sample from the beaker to the crucible. Dry the crucible with residue in an oven at 100 ±3 degrees Celsius (°C) for not less than 30 minutes.

c. Calculations:

A = sample weight

B = weight of crucible

C = weight of crucible plus residue composed of PTFE and magnesium (Mg)

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D = percent residue (PTFE and Mg)

E = percent copolymer

[(C -B)/A] X 100 = D

100 percent -D = E

Percent copolymer shall meet the requirements of 3.2.3.

d. Transfer the residue from the crucible with a stream of distilled water into a 250-ml glass joint flask. Add concentrated hydrochloric acid (HCl) to make a 2 to 3 normal (N) acid solution. Connect the flask to the condenser reflux apparatus and allow the sample to reflux until appearance of metal has been removed.

e. Allow the original crucible to set with a solution of 2 to 3 N HCl in order to remove any remaining metal. Wash the crucible well with distilled water and then wash with acetone to facilitate drying.

f. Transfer the sample of 4.4.3d into the original crucible. Dry the sample in an oven at 100 ±3°C for not less than 30 minutes. Cool the sample to room temperature in a desiccator and reweigh. Poly-TFE will be the final residue.

g. Calculations:

A = sample weight

B = weight of crucible

C = weight of crucible plus residue (PTFE and Mg)

F = weight of crucible plus residue (PTFE)

G = percent Mg

H = percent residue PTFE

[(C -F)/A] X 100 = G

[(F -B)/A] X 100 = H

Percent Mg (see G) and percent molding plastic (see H) shall meet the requirements of 3.2.3.

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4.4.4 Heat of explosion. The grain shall be tested in accordance with MIL-STD-286. The grain heat of explosion shall meet the requirements of 3.2.4.

4.4.5 Burning rate.

4.4.5.1 Test apparatus. Test apparatus shall consist of the following:

a. A burning rate apparatus designed to burn propellant strands underwater at constant pressure. The burning rate apparatus shall limit the pressure rise during burning to be not greater than 50 pounds per square inch (psi).

b. A timing instrument suitable for recording time for not less than 60 seconds and graduated to not less than 0.001-second increments.

c. A pressure gage to measure pressure not greater than 1,500 psi and graduated into 5-psi increments.

4.4.5.2 Ingredients. Ingredients shall consist of the following:

a. Nitrogen gas which shall contain not greater than 0.10 percent oxygen.

b. Polyvinyl chloride - acetate copolymer (Bakelite VYLF, Carbide and Carbon Chemicals Division, Union Carbide and Carbon Corporation, or equal).

c. Liquid plasticizer (Flexol R-2H, Carbide and Carbon Chemicals Division, Union Carbide and Carbon Corporation, or equal).

d. Methylene chloride, technical grade.

4.4.5.3 Inhibiting solution formulation. Inhibiting solution formulation shall be as specified in Table V (see 6.5).

TABLE V. Inhibiting solution formulation.

Ingredient	Amount
Methylene chloride	1950 ml
Polyvinyl chloride	360 grams
Liquid plasticizer	45 grams

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4.4.5.4 Burning rate determination. Burning rate determination shall be as follows:

a. Prepare 20 propellant strands for burning rate determination by cutting the edges of the grain and drilling holes for the ignition and timing wires as specified on Figure 2.

b. Insert the ignition wire through the ignition hole on the strand. Apply a small amount of the inhibiting solution, as prepared in accordance with 4.4.5.3, to both sides of the ignition hole to prevent water from dissipating the heat during the ignition process.

c. Insert the timing wires through the timing holes in the strand and assemble the strand into the strand burning apparatus as specified on Figure 3.

d. Burn five strands underwater at each of the following pressures: 500 psi, 700 psi, 800 psi, and 1,000 psi. The water temperature at the start of ignition shall be 24 ±3°C. The burning rate at each of these pressures shall be recorded. To be acceptable, the burning rate shall meet the requirements of 3.2.5.

4.5 Visual inspection. The grain shall be visually inspected and shall meet the requirements of 3.5.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level A.

5.1.1 Level A packaging. Grains shall be packaged in materials suitable for explosives and in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity. Packaging shall be in accordance with 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level A.

5.2.1 Level A packing. Grains, packaged in accordance with 5.1.1, shall be packed in materials suitable for explosives and in a manner that will provide protection from damage during direct shipment from the supply source to the first receiving activity. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 171-178.

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5.3 Marking.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with 49 CFR 171-178.

5.3.2 Normal markings. In addition to special markings required by 5.3.1 and the contract or purchase order (see 6.2.1), unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The grains are intended for use in MARK 264 and MARK 297 MOD 0 igniters.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

c. Special markings, if required (see 5.3.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.2.1.2	Test reports	DI-T-2072	---

(DIDs related to this specification and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 70 grains. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Definitions.

6.3.1 Batch. A batch of material is defined as that quantity of formulated material that has been subjected to one or more chemical or physical processes to produce a product having uniform characteristics.

6.4 Preparation of inhibiting solution. The following method for preparing the inhibiting solution is suggested. Other methods may be used, subject to approval by the contracting activity. Prepare the inhibiting solution as follows:

a. Add plasticizer to methylene chloride while agitating solution in stainless steel mix pot. Agitation is performed with a mechanical stirrer.

b. Slowly add vinyl plastic with constant agitation until evenly dispersed.

c. Pour into stock bottle and seal until ready for use.

6.5 Safety precautions. The handling, assembly, and loading of the grains covered by this specification involve hazardous materials, and therefore require explosive safety precautions. The standard safety precautions for explosive-loaded items contained in DOD 4145.26M, "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" are applicable. When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of OP 5, "Ammunition Ashore" are applicable.

6.6 Subject term (key word) listing.

Grain
Igniter
Manesium
Polytetrafluoroethylene (PTFE)
Rocket motor

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6.7 Suggested source of supply. A product that has met the requirements of the this specification for plastic molding material (PTFE), Table II, in past procurement actions is Haylon TFE G-10 manufactured by Allied Engineered Plastics, Allied Corporation, P.O. Box 2332R, Morristown, New Jersey 07960, Code Ident 70308. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Preparing activity:

NAVY - AS
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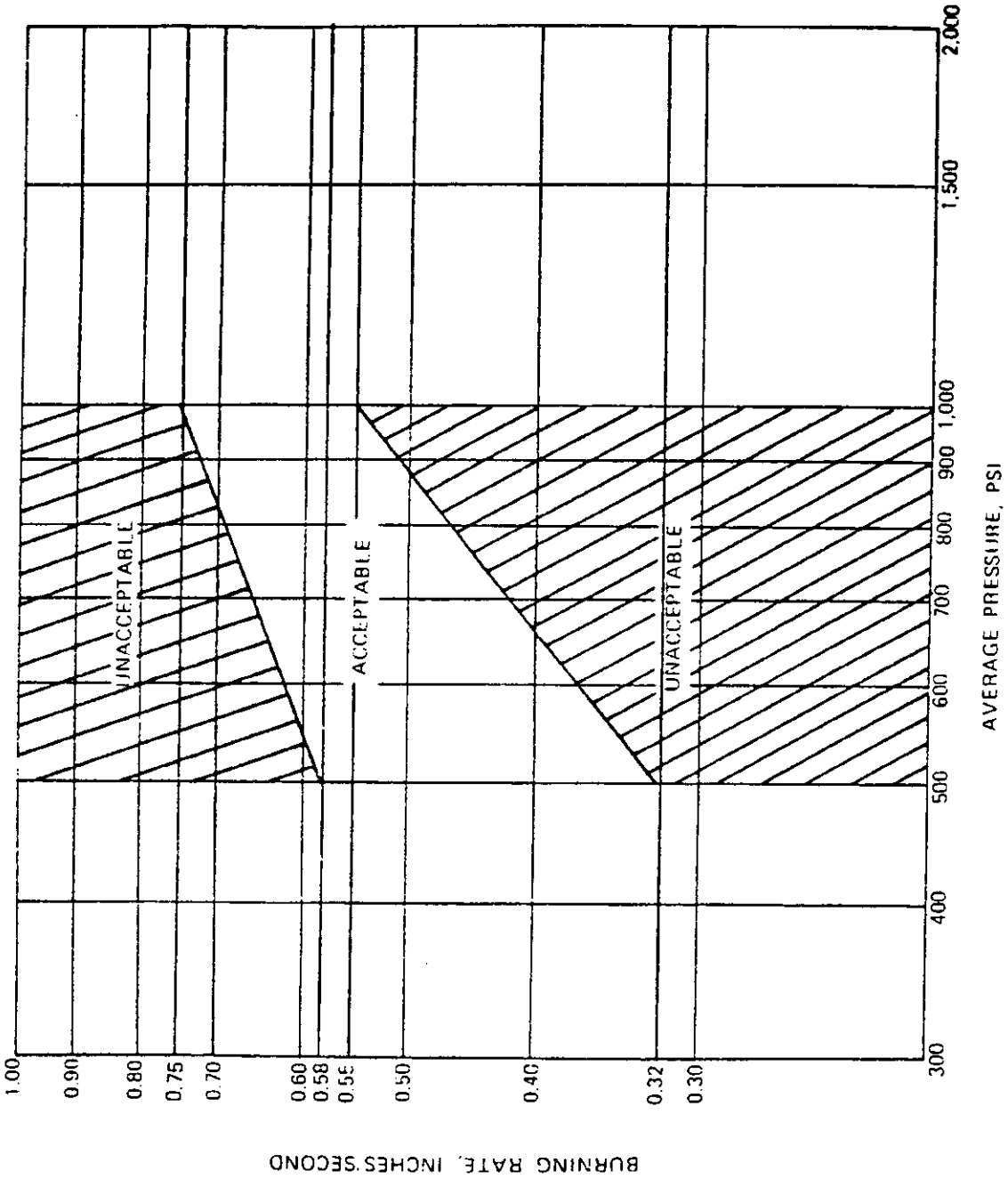


FIGURE 1. Burning-rate limits.

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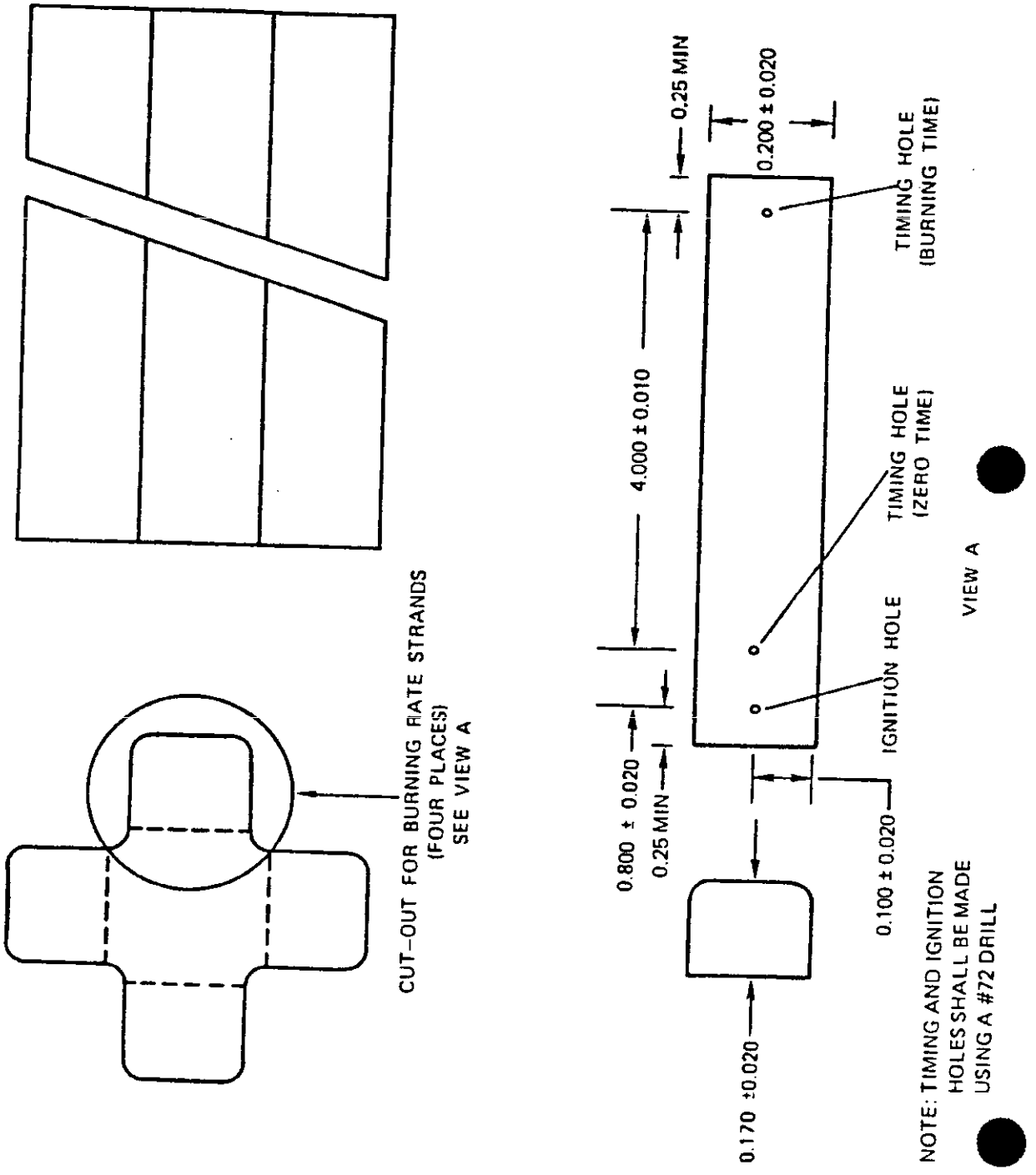


FIGURE 2. Burning-rate strands preparation.

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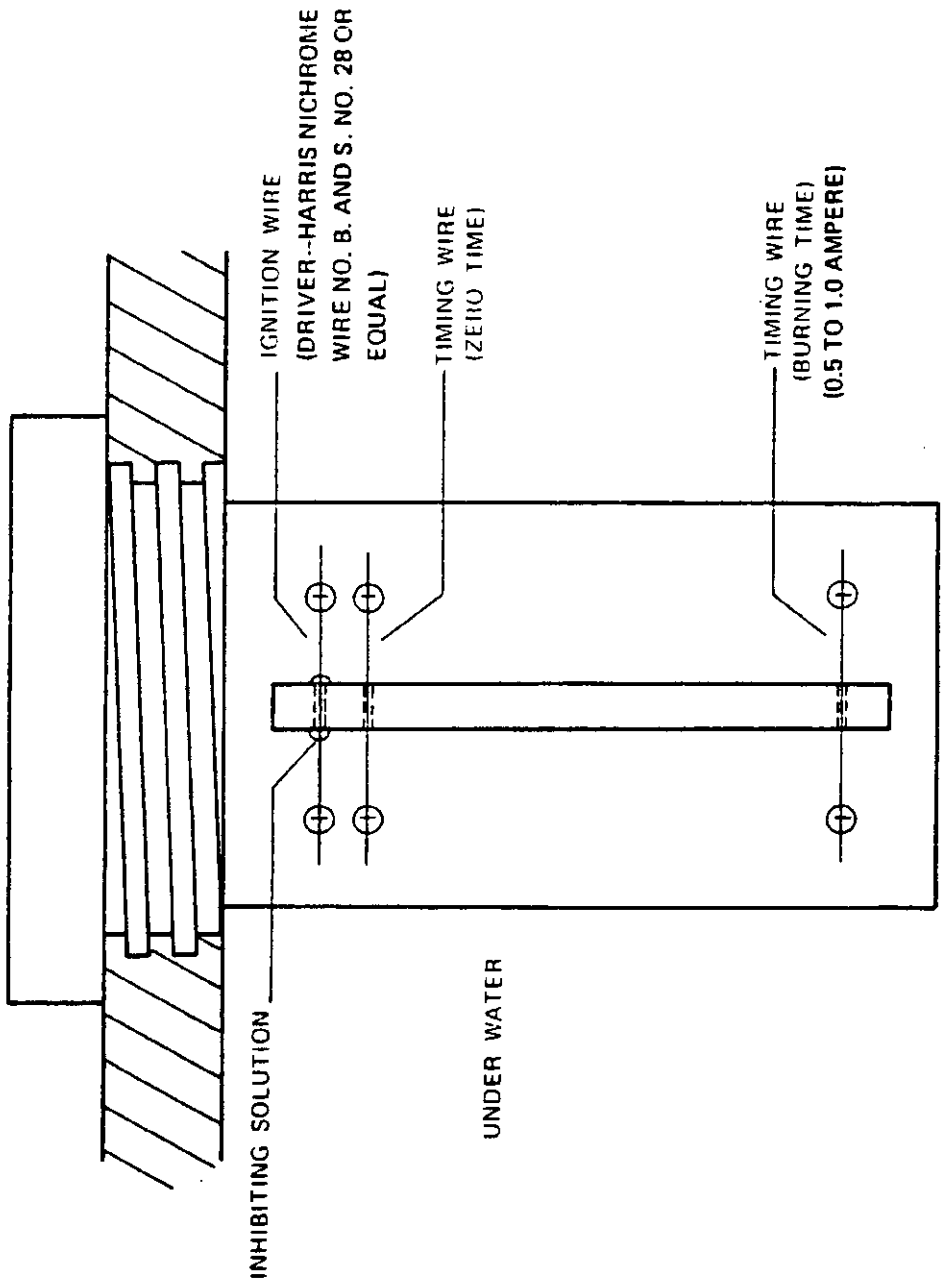


FIGURE 3. Typical underwater burning assembly.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-85719(AS)	GRAIN, IGNITER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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